GlucoTwin

A hybrid digital twin that warns of glucose spikes and lows in Type 2 Diabetes before they happen, by fusing EHR, CGM and wearable data, validated on real patients

<Team Name> · <College / Incubator>

Digital Twin Challenge 2026 · Happiest Health · Reimagining & Reforming Healthcare in India Summit

THE PROBLEM

Diabetes in India is managed by looking backwards

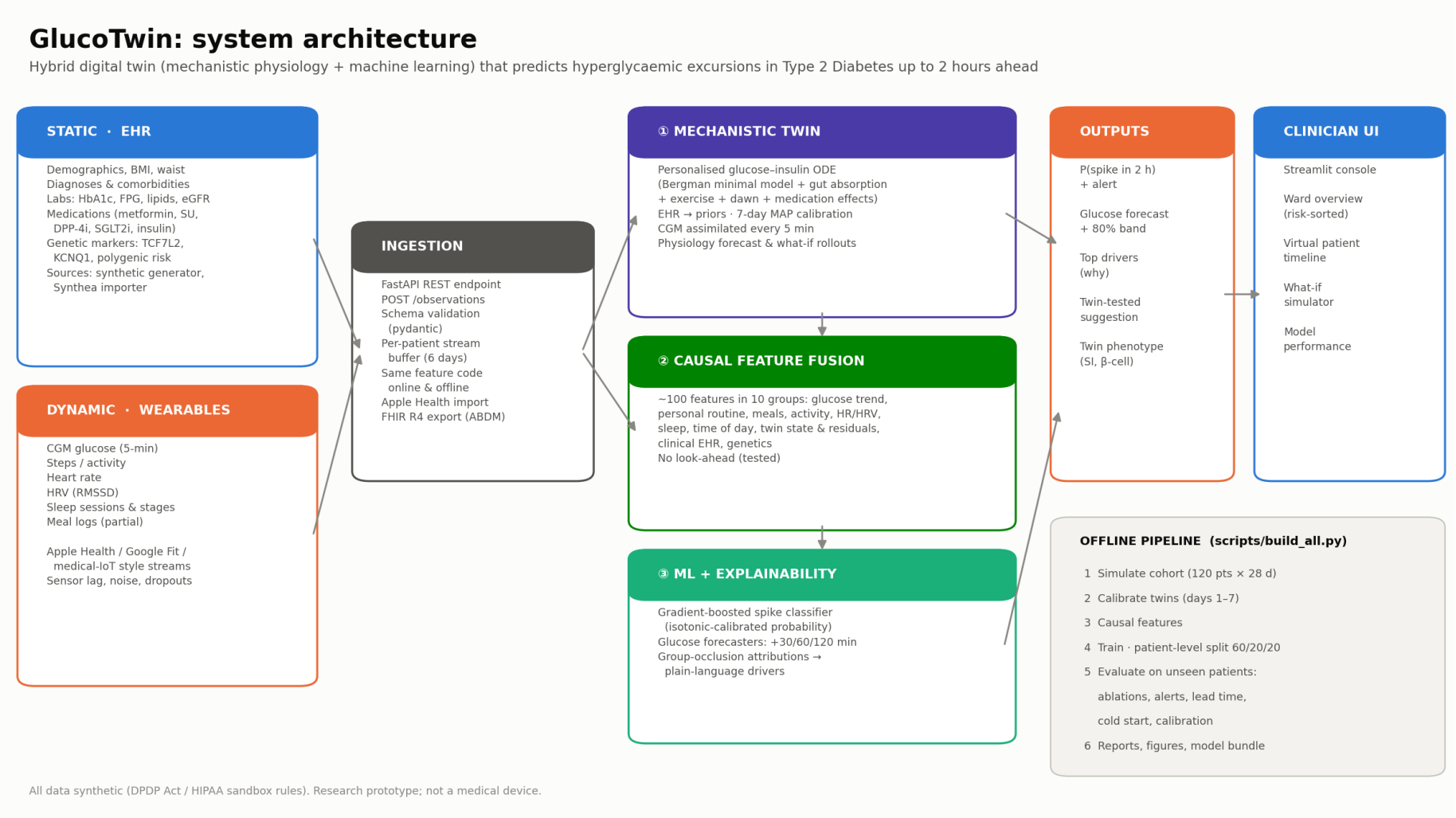
- India has ~101 million adults with diabetes and ~136 million with prediabetes (ICMR-INDIAB, 2023).
- High-carbohydrate meals (rice, roti, dosa, biryani) drive sharp post-meal glucose excursions, linked to vascular damage even when HbA1c looks acceptable.
- HbA1c is measured every ~3 months: it averages away the daily spikes.
- CGMs show what already happened, not what is about to happen.
- Doctors see a patient for minutes every few months, with no way to test 'what if this patient walked after dinner?'

Our question

Can a personalised virtual replica of a patient, continuously updated from wearables and anchored in their health record, warn of a glucose spike early enough to act, and show which action would help?

OUR SOLUTION

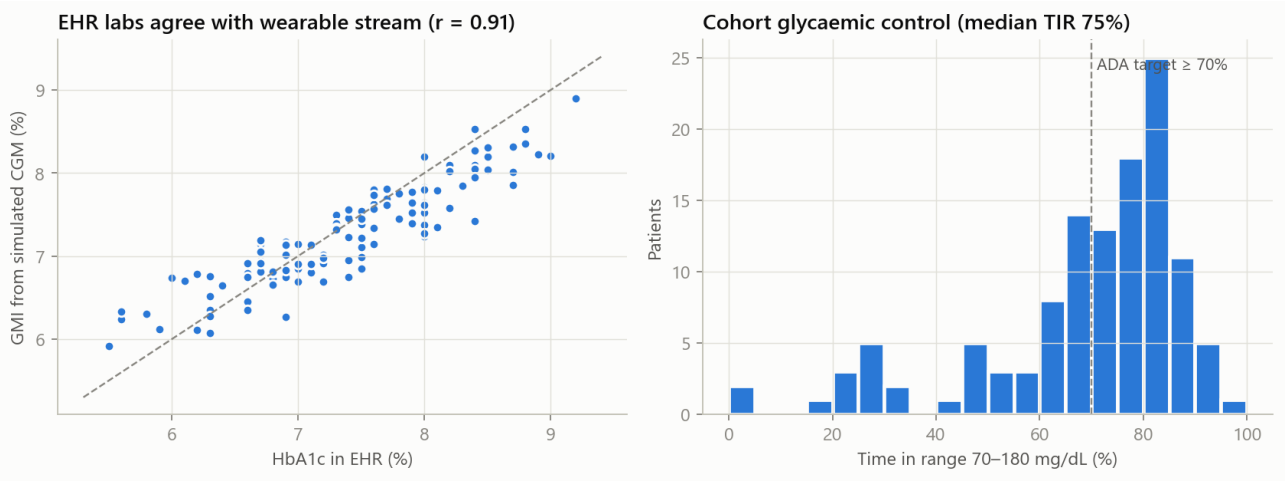
A three-layer hybrid digital twin



DATA

Sandbox-compliant data: a causally consistent synthetic cohort

- 120 synthetic Indian T2D patients × 28 days at 5-minute resolution (≈1M rows).
- Static EHR: demographics, labs, comorbidities, 6 drug classes, TCF7L2/KCNQ1 risk alleles, polygenic score.
- Dynamic streams: CGM, steps, heart rate, HRV, sleep stages, partial meal logs.
- Generated causally: genetics + meds → physiology → glucose → sensors and labs, so HbA1c agrees with CGM.
- Hidden drivers the model must infer: unlogged meals (≈1 in 3), stress (HR↑ HRV↓), sleep debt.
- Realistic noise: CGM lag, ~7% MARD, dropouts, sensor change, HRV gaps.
- Real-data path built and tested: Synthea EHR importer, Apple Health importer, FHIR R4 export.

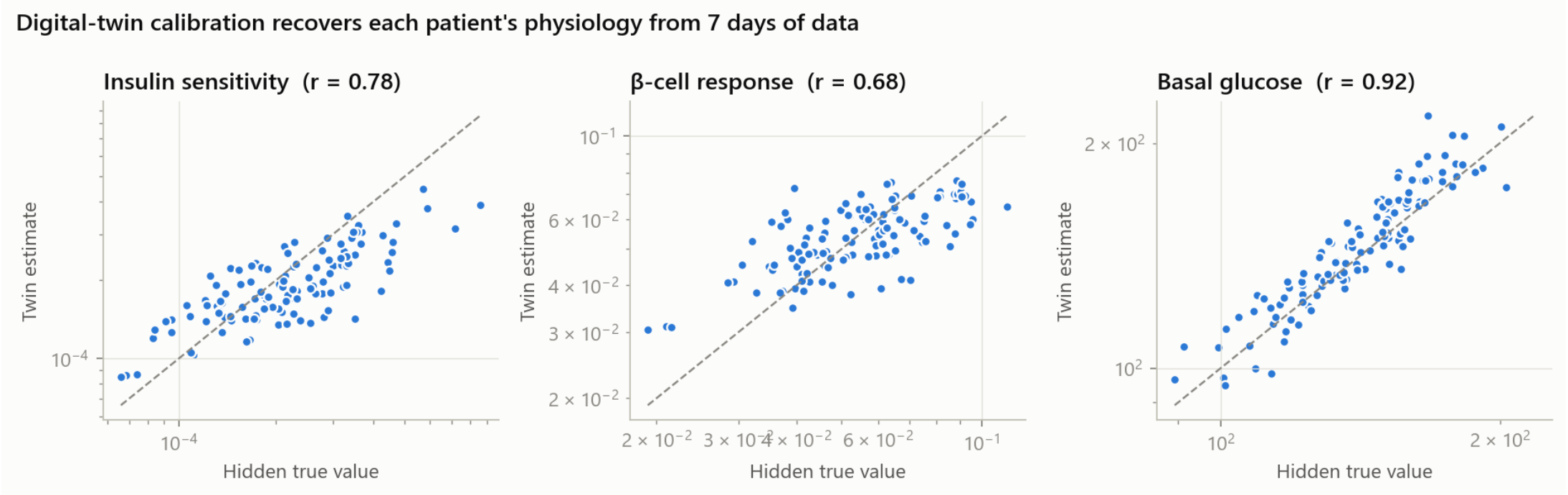


MECHANISTIC MODEL

Layer ①: a personalised physiological twin

$dG/dt = -(SG + X \cdot m) \cdot G + SG \cdot Gb + Ra(meals) + dawn - exercise - SGLT2\ excretion$ · $dX/dt = -p2 \cdot X + p2 \cdot SI \cdot (I + I_exo)$ · $dI/dt = -n \cdot I + \beta \cdot (G - Gb)^{+} + s0$

EHR sets the priors (weight, FPG, BMI, duration, medications). One onboarding week of CGM + meals + steps personalises insulin sensitivity, β-cell response, basal glucose and dawn effect (MAP fit). CGM is assimilated every 5 minutes.



PERFORMANCE

Results on 24 patients the model never saw

0.952

AUROC for predicting a spike within the next 2 hours

82%

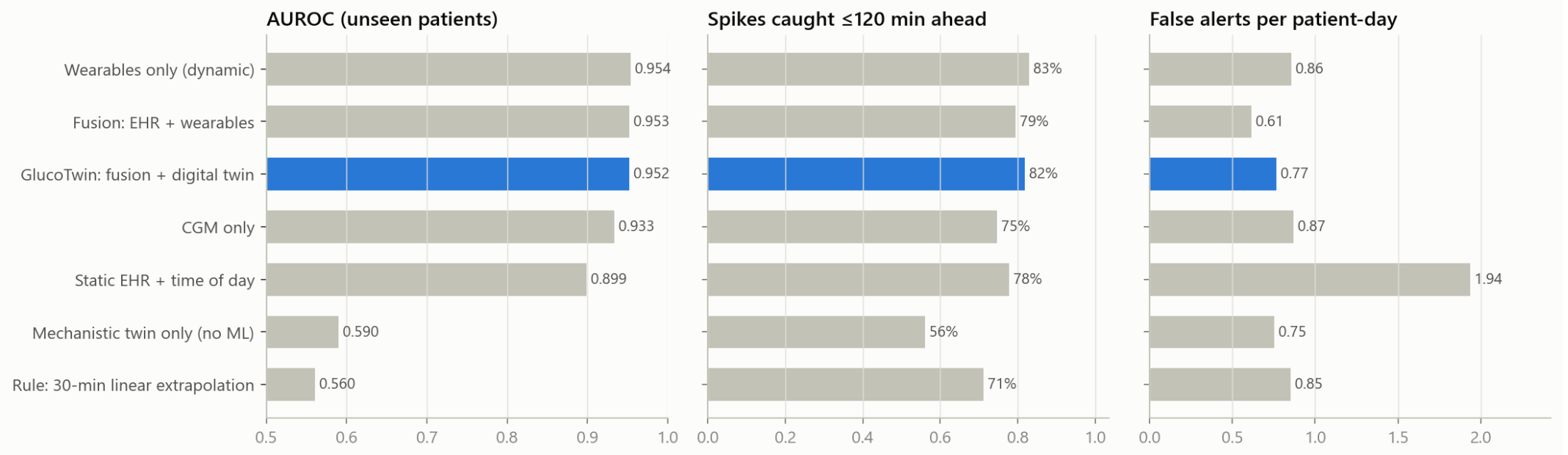
of spikes warned about in advance

75 min

median warning time before the spike starts

0.77

false alerts per patient-day

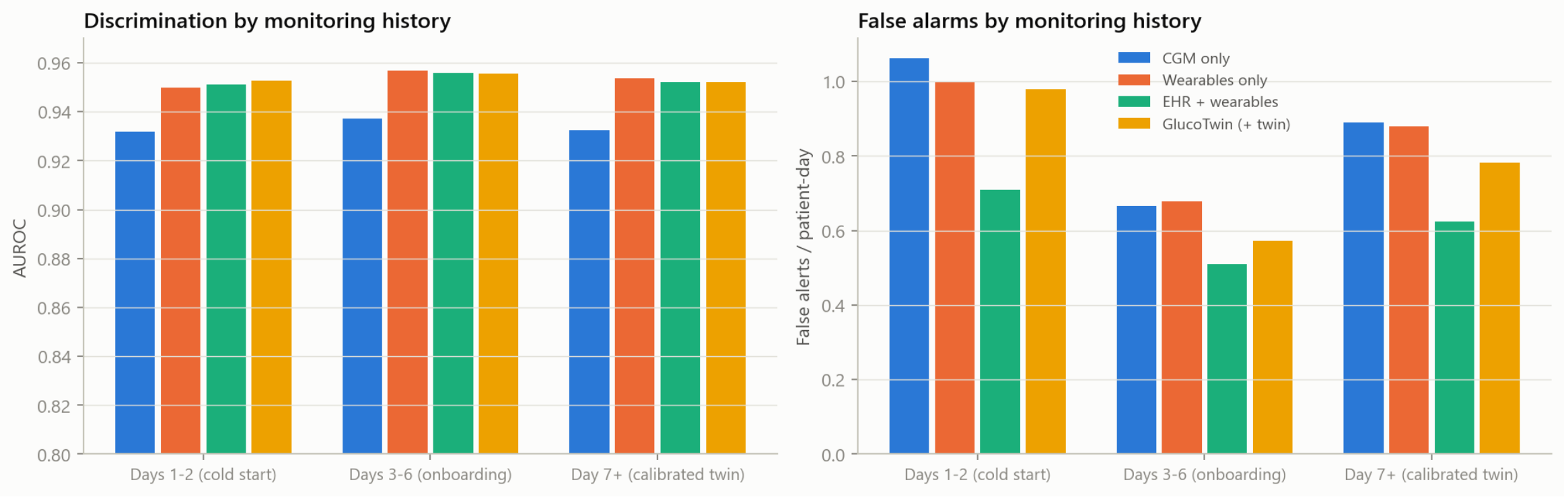


ABLATION

Why fuse? Each data stream earns its place

• CGM alone: AUROC 0.933, 0.87 false alerts/day. Adding meals, activity, sleep, HR/HRV and personal routine: 0.954 and 0.86/day (AUROC gain +0.020, 95% CI +0.013 to +0.026).

• Once wearable history exists, static EHR and twin features add no measurable discrimination (differences within ±0.002). The twin's role is simulation and explanation, not raw accuracy.

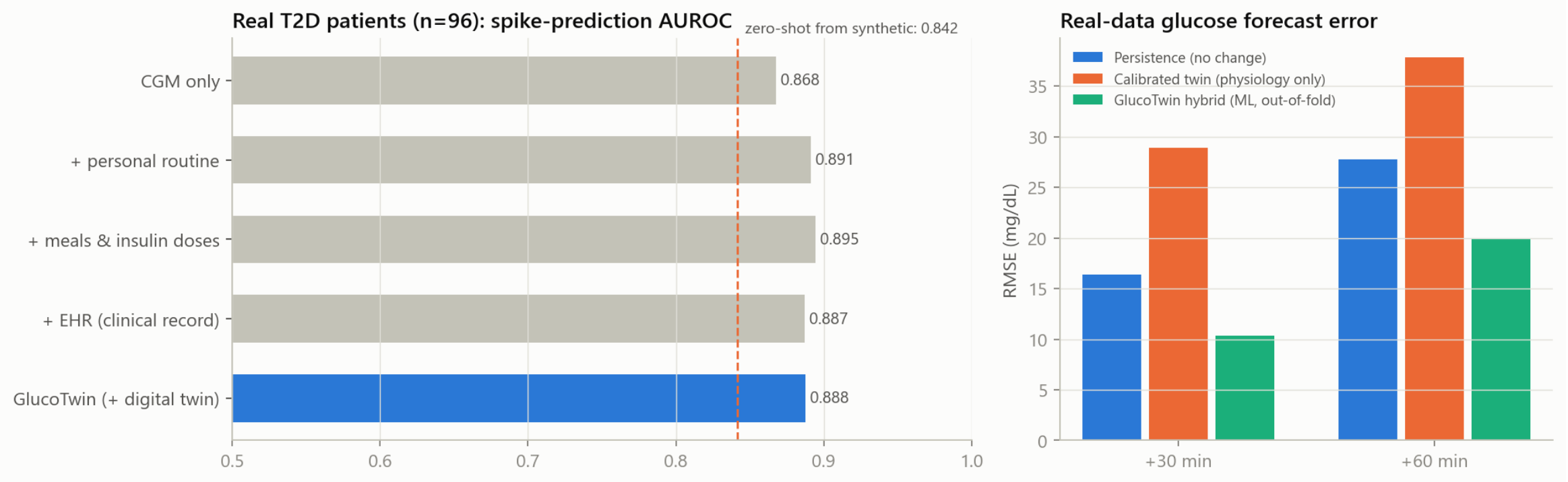


REAL-WORLD EVIDENCE

Validated on 96 real people with type 2 diabetes

ShanghaiT2DM (Sci Data 2023, CC BY 4.0): CGM + clinical record + meals + insulin doses. 5-fold cross-validation grouped by person.

Fusion beats CGM alone (+0.020 AUROC, 95% CI +0.012 to +0.031). Trained only on synthetic data: 0.842. Honest negatives: static EHR slightly hurt; the physics-only twin forecast is worse than 'no change' on these patients.

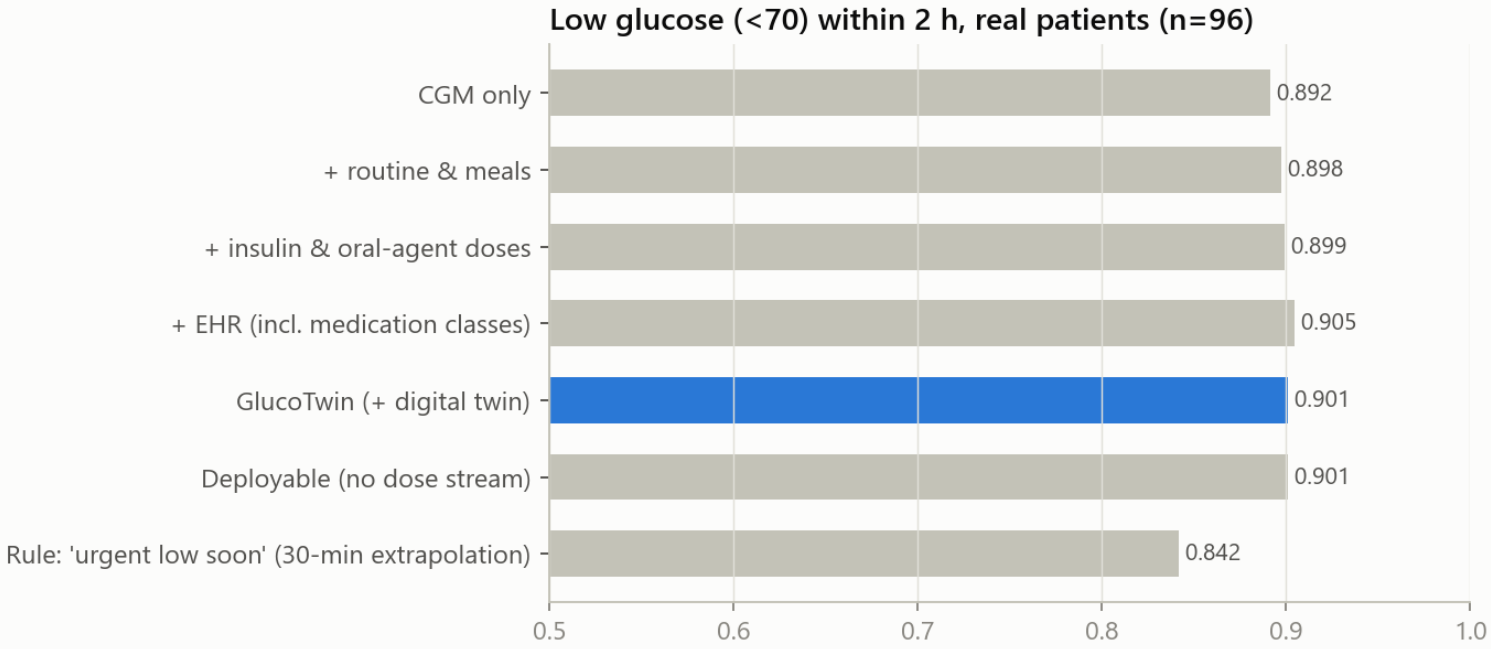


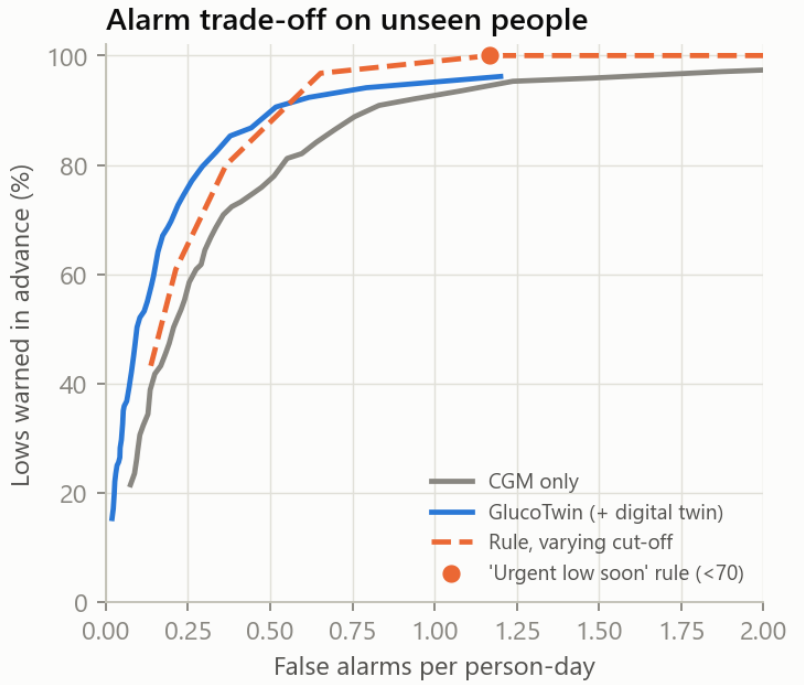
HYPOGLYCAEMIA

Both directions of danger: predicting lows on real patients

Lows (<70 mg/dL) are the dangerous side of treatment, especially with insulin or sulfonylureas. Trained and validated on 96 real people (343 low episodes), with person-grouped cross-validation.

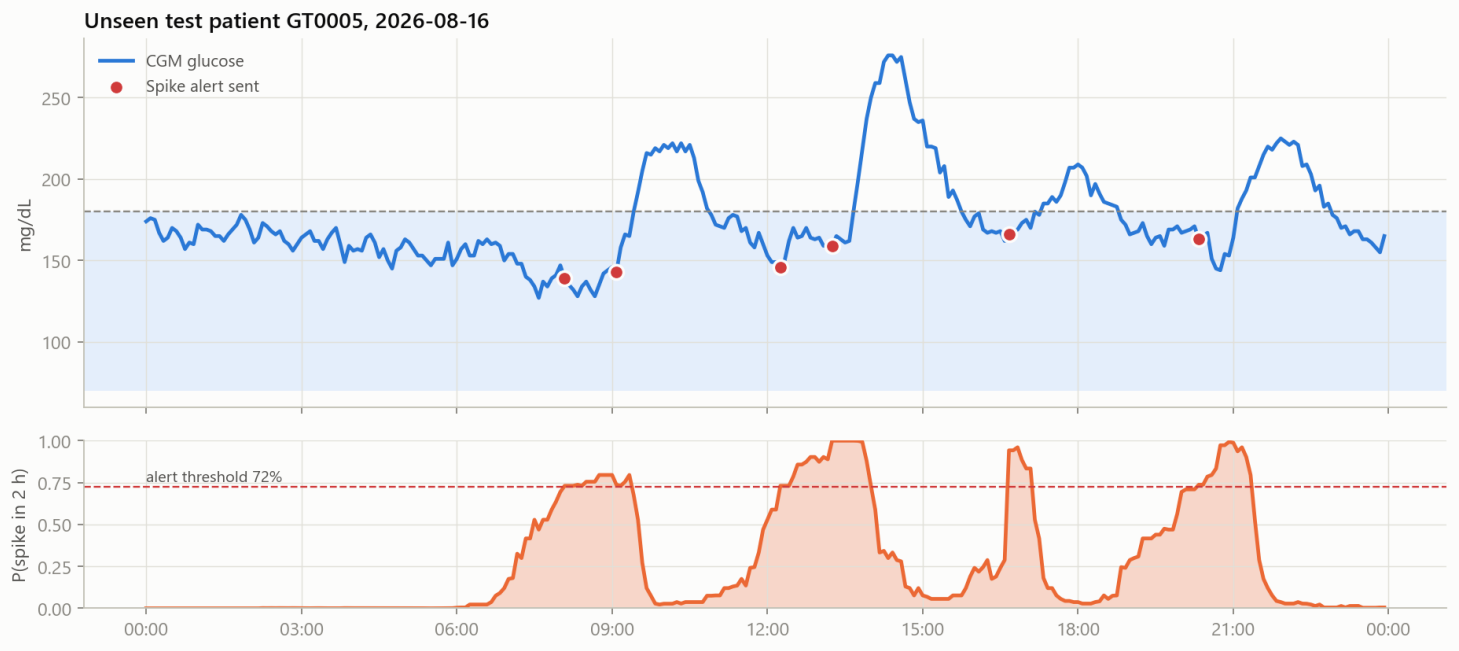
GlucoTwin AUROC 0.901 vs 0.842 for the 'urgent low soon' rule that CGMs use. Mixed but useful: GlucoTwin catches more lows when alarms must be rare (alarm fatigue), the simple rule wins when alarms can be frequent, so we use both: the rule as a safety net, GlucoTwin for earlier, quieter warnings.

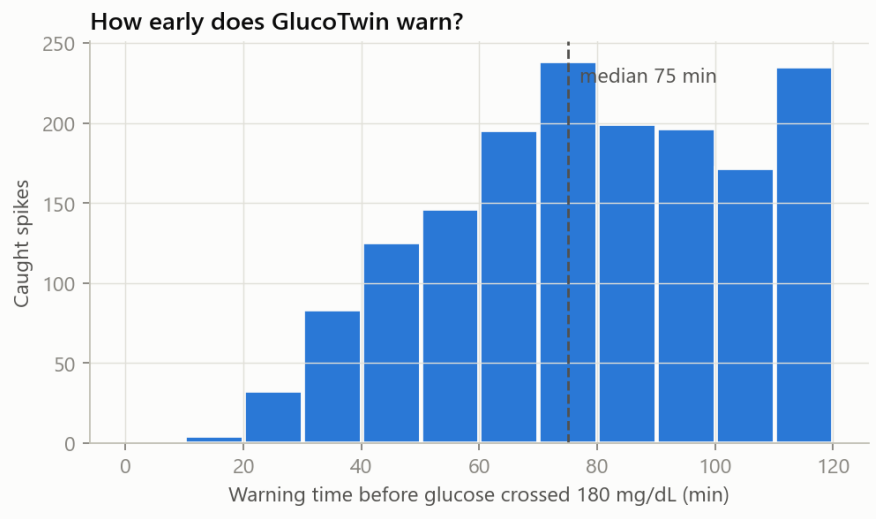




LEAD TIME

Early, actionable warnings

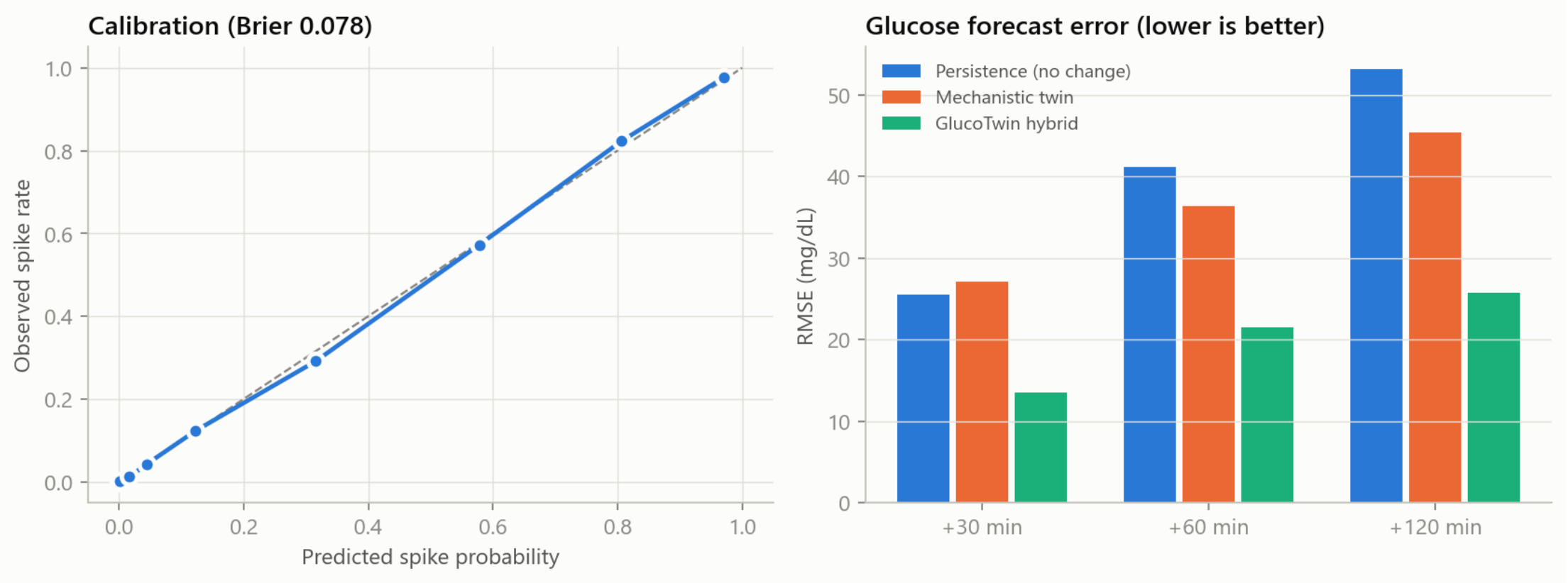




FORECASTING

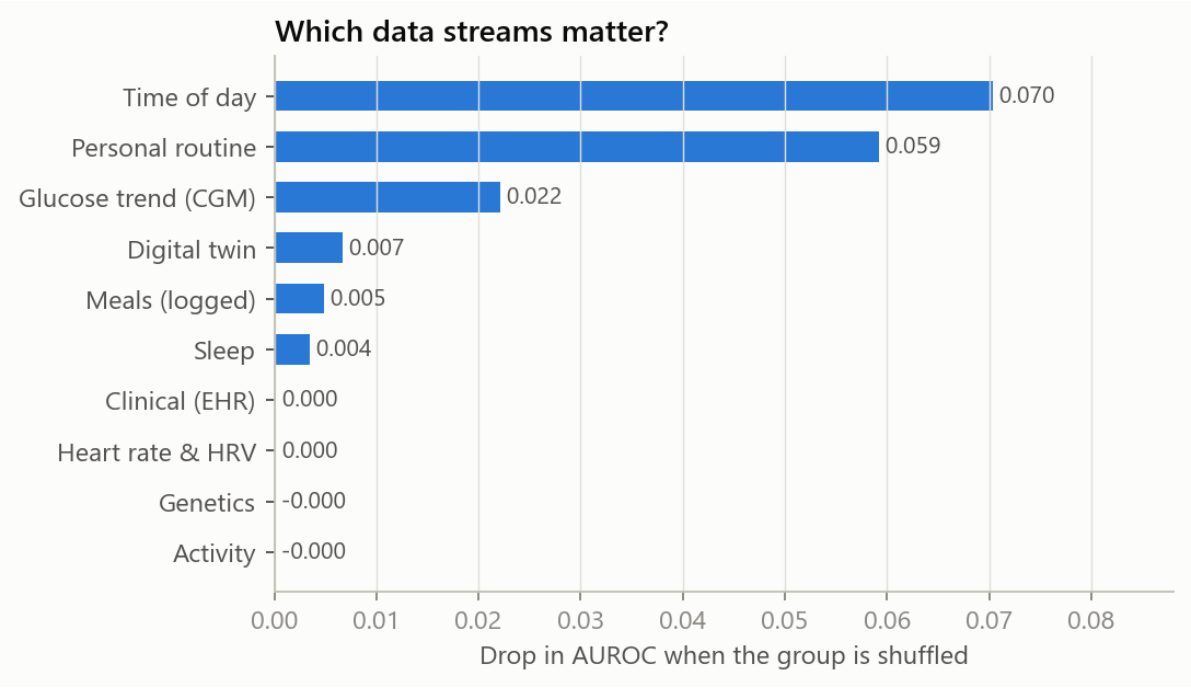
Trustworthy numbers: calibrated risk and accurate forecasts

Hybrid forecast error at +30 / +60 / +120 min: 13 / 21 / 26 mg/dL, roughly half the error of the physiology-only twin or a 'no change' guess.



WHY DID IT ALERT?

Explainable by design



Every alert comes with its top drivers, for example:

▲ Personal routine: spiked above 180 around this time on 3 of the last 3 days

▲ Meals: 75 g carbs logged 20 min ago

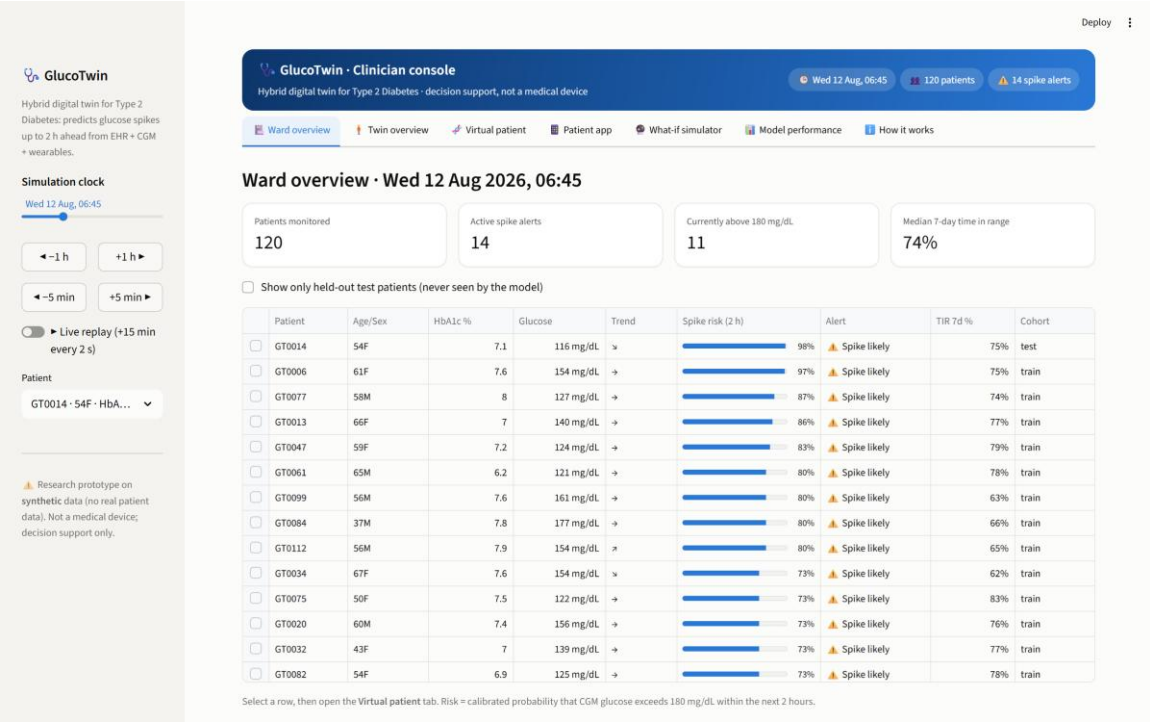
▲ Digital twin: glucose 28 mg/dL above twin expectation, possible unlogged meal

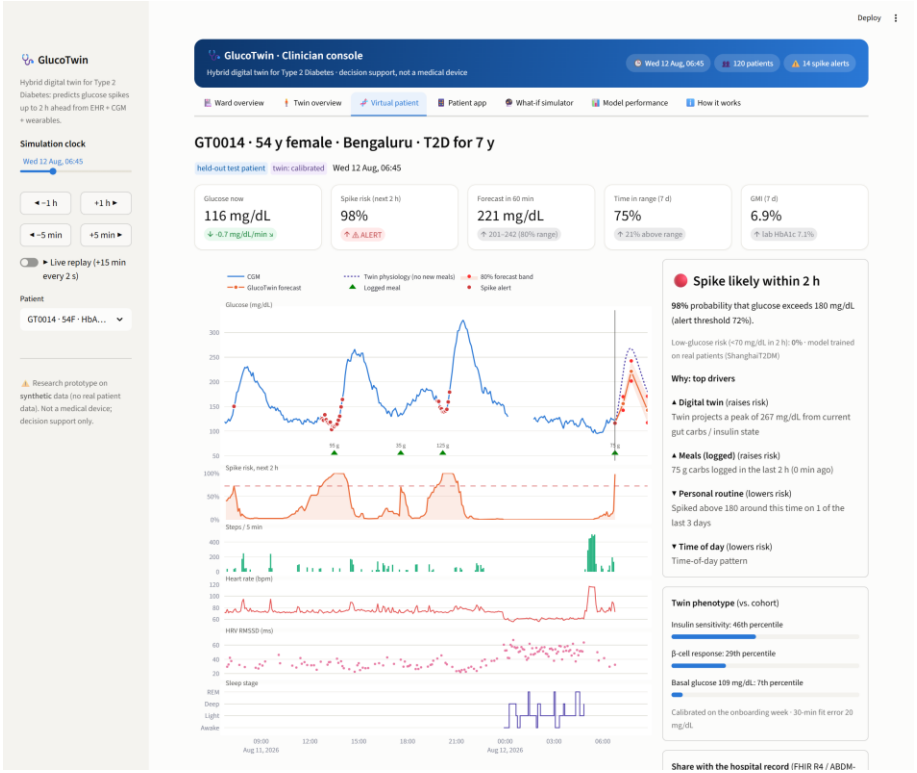
▲ Sleep: slept 4.9 h last night (−1.8 h vs usual)

…plus a twin-tested suggestion, e.g. 'a 15-minute brisk walk now lowers the predicted peak by ~20 mg/dL'.

CLINICIAN DASHBOARD

How a doctor uses the twin

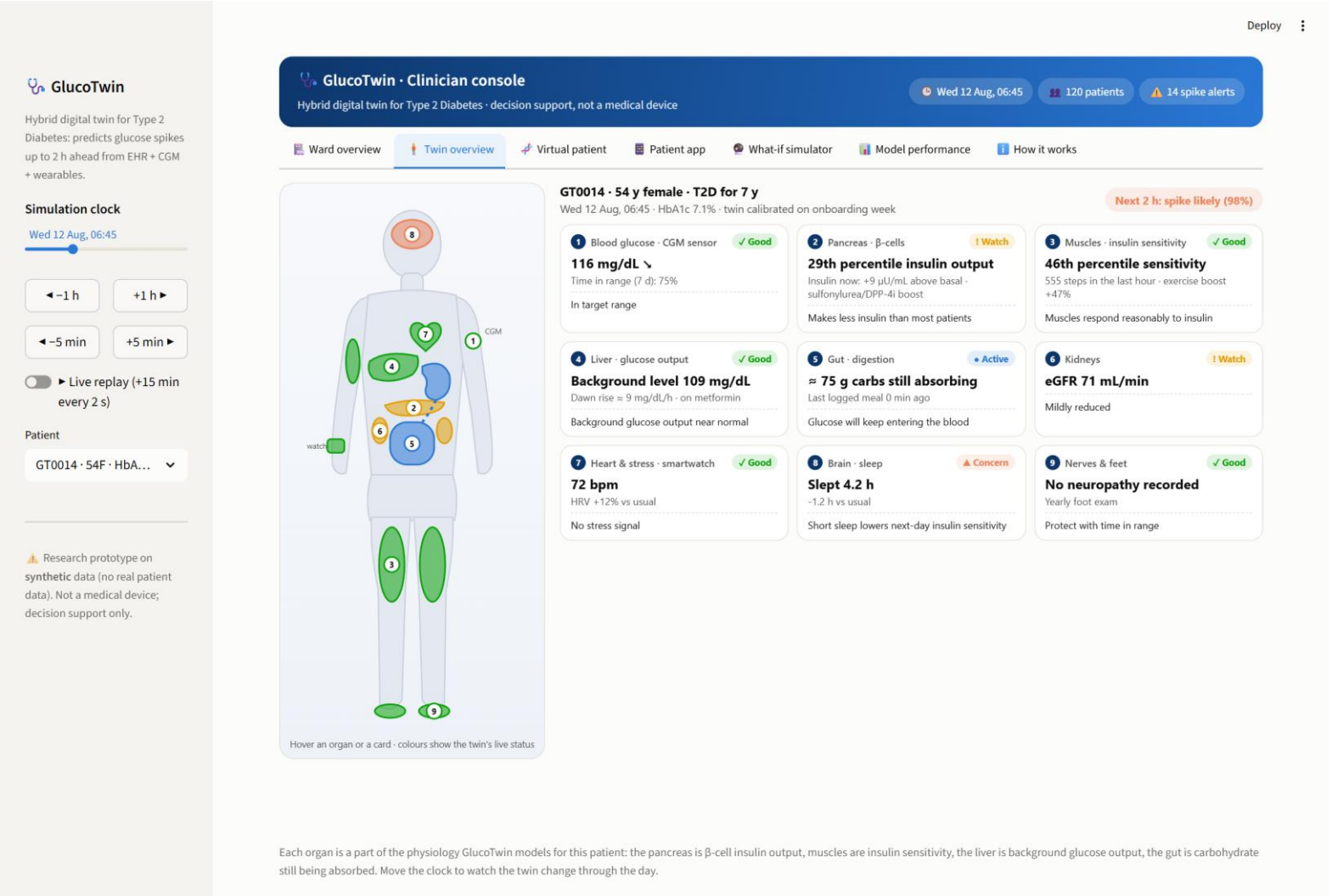




Ward overview (risk-sorted) → Virtual patient (CGM, forecast, risk, activity, HR/HRV, sleep, EHR, AGP, twin phenotype) → What-if simulator (Indian meal presets, carb quality, post-meal walk, sleep) → Model performance

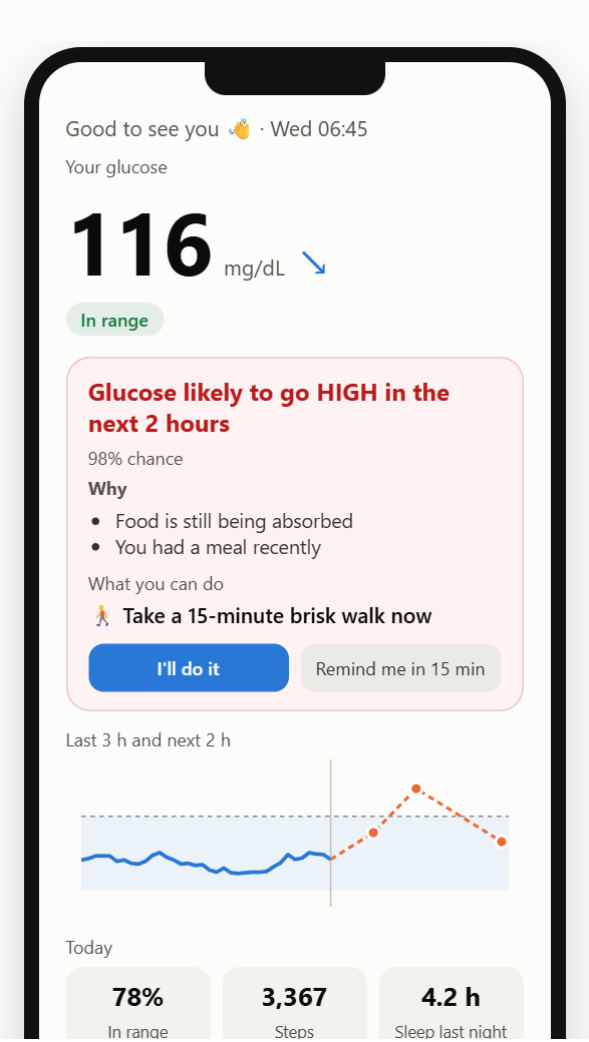
TWIN OVERVIEW

The twin at a glance: the patient's body, live



PATIENT APP

What the patient sees: one number, one reason, one action



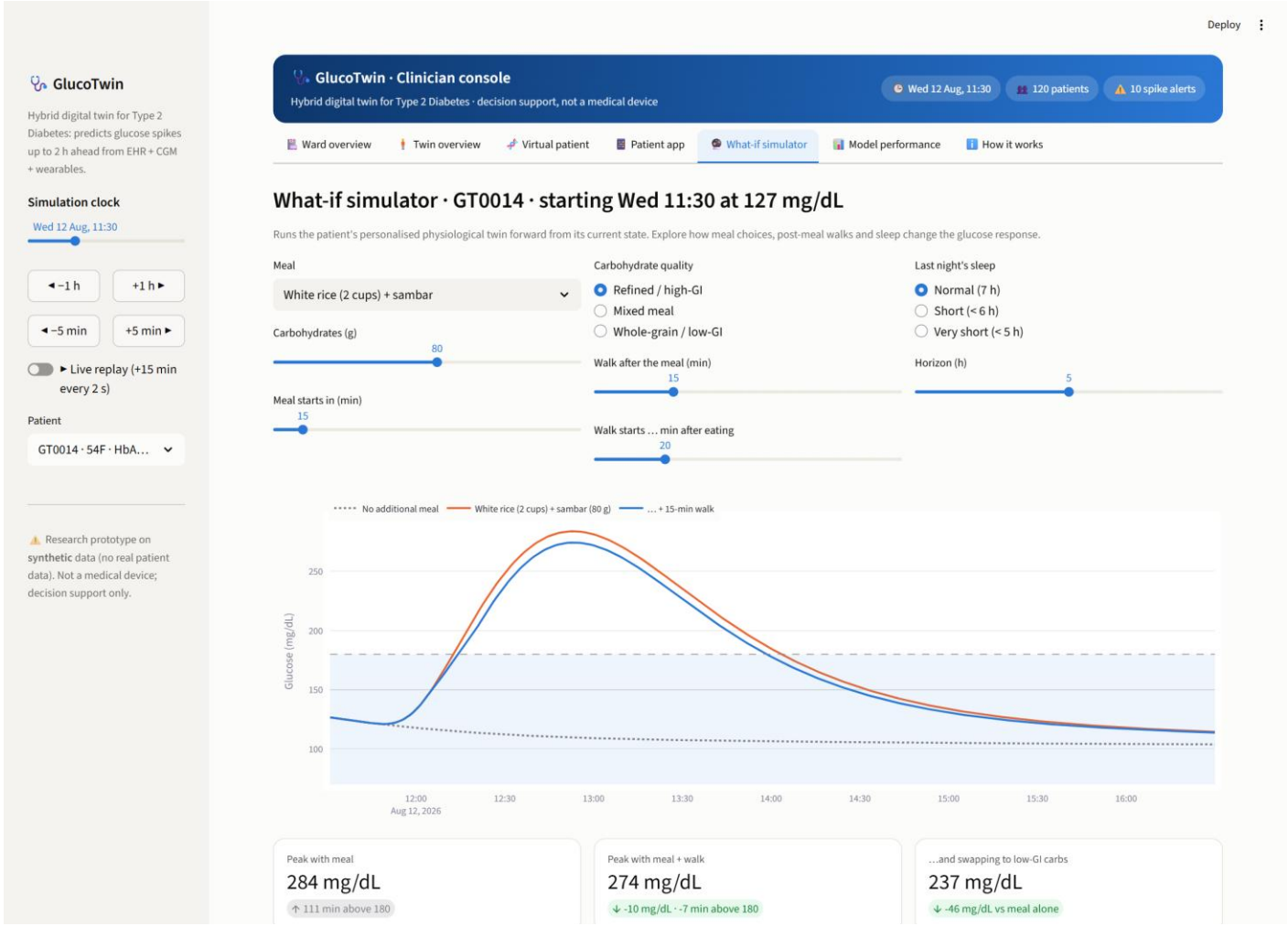
The clinician gets the full console; the patient gets a phone screen:

- glucose now + trend
- the prediction in plain words (English or Hindi)
- the top reason, e.g. 'You had a meal recently'
- one action the twin has tested: a 15-minute walk
- lows take priority over highs

The 'I'll do it / Remind me' taps would feed adherence back to the care team.

SIMULATION

What-if: personalised treatment exploration



The calibrated twin is run forward from the patient's current state:

- Which meal? (rice + sambar vs ragi mudde)
- Walk after dinner? How long, how soon?
- Effect of a short night's sleep

For counselling and shared decision-making, not dosing.

ENGINEERING

Built like a product, not a notebook

- Real-time engine + FastAPI: POST observations → GET assessment / what-if
- Replay script streams a patient like a device gateway
- Identical feature code online and offline (tested to match)
- Leakage test: changing the future never changes today's features
- Patient-level train/val/test split; thresholds chosen on validation patients
- Alerts evaluated as real notifications (re-sent hourly while risk stays high)
- One command rebuilds everything: python scripts/build_all.py
- New-patient lifecycle over the API: register → stream → calibrate → FHIR RiskAssessment
- docker compose up runs dashboard + API

Stack

Python · NumPy · pandas · SciPy · Numba (ODE kernels) · scikit-learn (HistGradientBoosting, isotonic calibration) · FastAPI · Streamlit · Plotly · pytest

License: MIT

HONEST ASSESSMENT

Limitations and next steps

Limitations

- Main results are synthetic; the real-data check (96 Chinese T2D patients, CGM but no other wearables) confirms the fusion gain but not the twin's forecasting.
- What-if simulations are not validated against real interventions.
- The low-glucose model is trained only on the real Shanghai cohort (lows are too rare in the simulation).
- The dawn-effect parameter is not identifiable from one week.
- Twin features add no measurable accuracy over the fused ML model, even when trained on only 6 patients; the twin's value is what-if simulation and interpretability.
- Once wearable history exists, static EHR adds little discrimination.

Next steps

- Prospective Indian cohort with CGM + smartwatch (steps, HR, HRV, sleep) to test the full fusion.
- Connect the FHIR R4 export to an ABDM health-information gateway (export itself is built).
- Validate the low-glucose model on Indian patients and add nocturnal-low alerts.
- Prospective pilot with clinicians, then the CDSCO SaMD pathway.

Thank you

<Team Name> · <College / Incubator> · github.com/<your-org>/<repo>